

A Constant Prime Orbit for TLMC 00000000118

Prepared for the account `idealistic hacker`

Frozen statement and explicit reading

The organizer source at commit `95acb520ec5607c826b8a997b1ef2fc82d6f7c57` states in English:

“For every M there exists an integer-coefficient quadratic f such that $f(0), f^1(0), \dots, f^M(0)$ are all prime.”

The Chinese source was separately read and gives the same displayed condition. The frozen Git blob is `697131a7b8dc0a62c539018c7d22612df1e7909a`; the SHA-256 of its raw Git-blob bytes is `91b64b4a671db1806c346247acbc2e0d849c5f73337bc807806c7cd00bbd580a`.

We read superscripts as compositional iteration:

$$f^{\circ 0}(x) = x, \quad f^{\circ(n+1)}(x) = f(f^{\circ n}(x)).$$

The written list starts with $f(0)$, not $f^{\circ 0}(0) = 0$. Accordingly, the formal statement asks for $f(0)$ and for $f^{\circ i}(0)$ when $1 \leq i \leq M$. It does not require the nonprime number 0 to be prime. For $M = 0$, the resulting requirement is $f(0)$ alone. The source does not impose pairwise distinctness of the displayed values.

Construction

Set

$$f(x) = x^2 - 2x + 2 = (x - 1)^2 + 1.$$

This has integer coefficients and nonzero quadratic coefficient 1. Moreover,

$$f(0) = 2 \quad \text{and} \quad f(2) = 2.$$

Hence every positive iterate at zero is the same prime.

Theorem 1. *For every natural number M , the displayed prime condition holds for the integer-coefficient quadratic $f(x) = x^2 - 2x + 2$ under the explicit reading above.*

Proof. The initial displayed value is $f(0) = 2$, which is prime. For each $i \geq 1$, write $i = n + 1$. Induction on n shows $f^{\circ(n+1)}(0) = 2$: the base case is $f(0) = 2$, and the induction step applies $f(2) = 2$. Thus every requested value is 2, so it is prime. The same f works for each M . $\square$

Lean correspondence and audit

The dependency-free Lean 4.33.1 project formalizes the construction in `lean4/Main.lean`. Its final theorem is

```
PrimeChain.iterated_prime_chains : forall M : Nat, HasPrimeChain M.
```

The predicate `IntegerQuadratic` explicitly exhibits integer coefficients and proves the leading coefficient nonzero. `HasPrimeChain` encodes the index range stated above. `lean4/Check.lean` prints the axiom footprint for every theorem advertised here. The proof does not rely on a finite computation in place of the quantified theorem.

Status and boundary

This PDF was compiled locally with cached-only Tectonic 0.15.0. The package has no independent review record and makes no claim of organizer approval, merge, acceptance, rank, award, or human verification. Independent statement-alignment review, mathematical review, Lean review, reproducibility review, and fresh release-coordinator checks remain mandatory before any publication.